

NAME	KEN KIBUGI MUIRURI
TRACKING NUMBER	CS-SA12-26014
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TASK	PASSIVE RECONNAISSANCE REPORT

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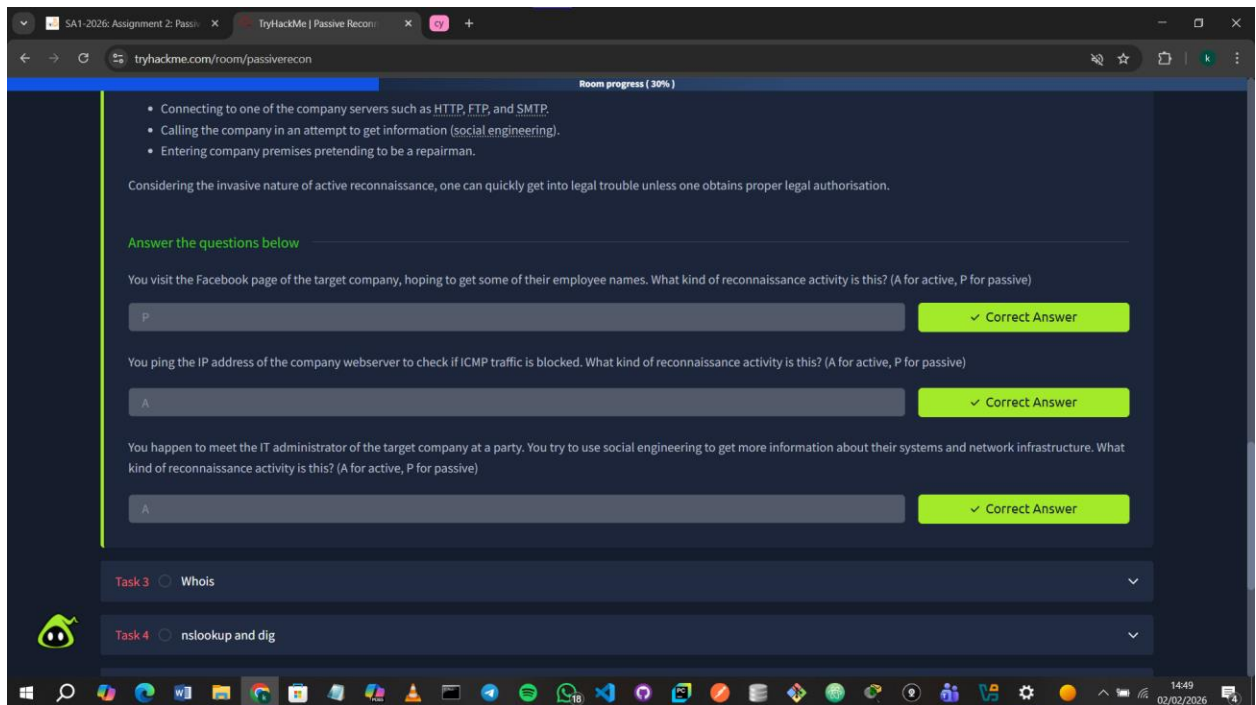
PASSIVE RECONNAISSANCE REPORT

Introduction

This report outlines what I learned from the TryHackMe Passive Reconnaissance module as part of my Cyber Shujaa training. The module focused on understanding how publicly available information can be gathered about a target without direct interaction. It introduced key tools and techniques used during the early stages of a security assessment.

1. Passive Versus Active Recon

In this section, I learned that reconnaissance is the initial stage of an attack where information is gathered about a target. Passive reconnaissance involves collecting publicly available information without directly interacting with the target, while active reconnaissance requires direct engagement with systems or people.



2. Whois

In this section, I learned that WHOIS is a protocol used to retrieve publicly available information about a domain, such as the registrar, registration dates, contact details, and name servers. I also learned how to use the whois command in Linux to gather this information during passive reconnaissance. This data can help identify potential technical or social engineering attack surfaces, although privacy services may limit what is visible.

I got the answers to the questions by running **whois tryhackme.com** on my kali vm

Question1: When was TryHackMe.com registered?

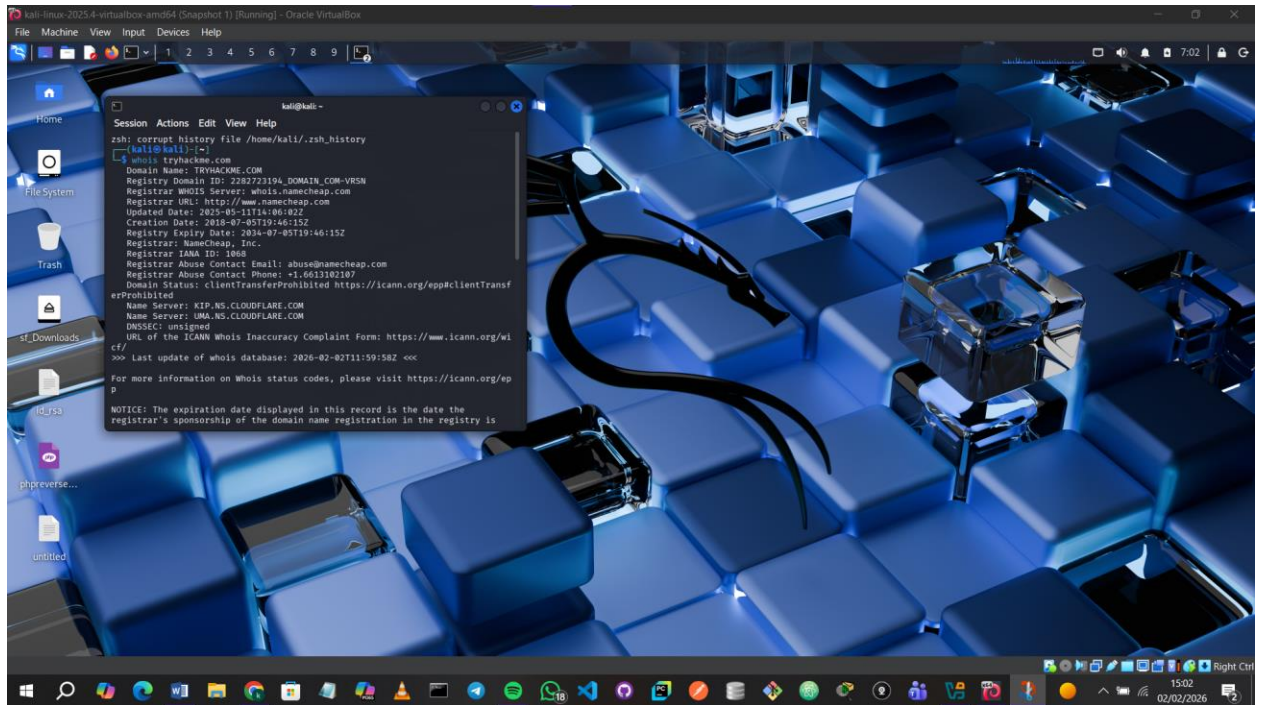
Answer: **20180705**

Question2: What is the registrar of TryhackMe.com?

Answer: **namecheap.com**

Question3: Which company is TryHackMe.com using for name services?

Answer: **cloudflare.com**

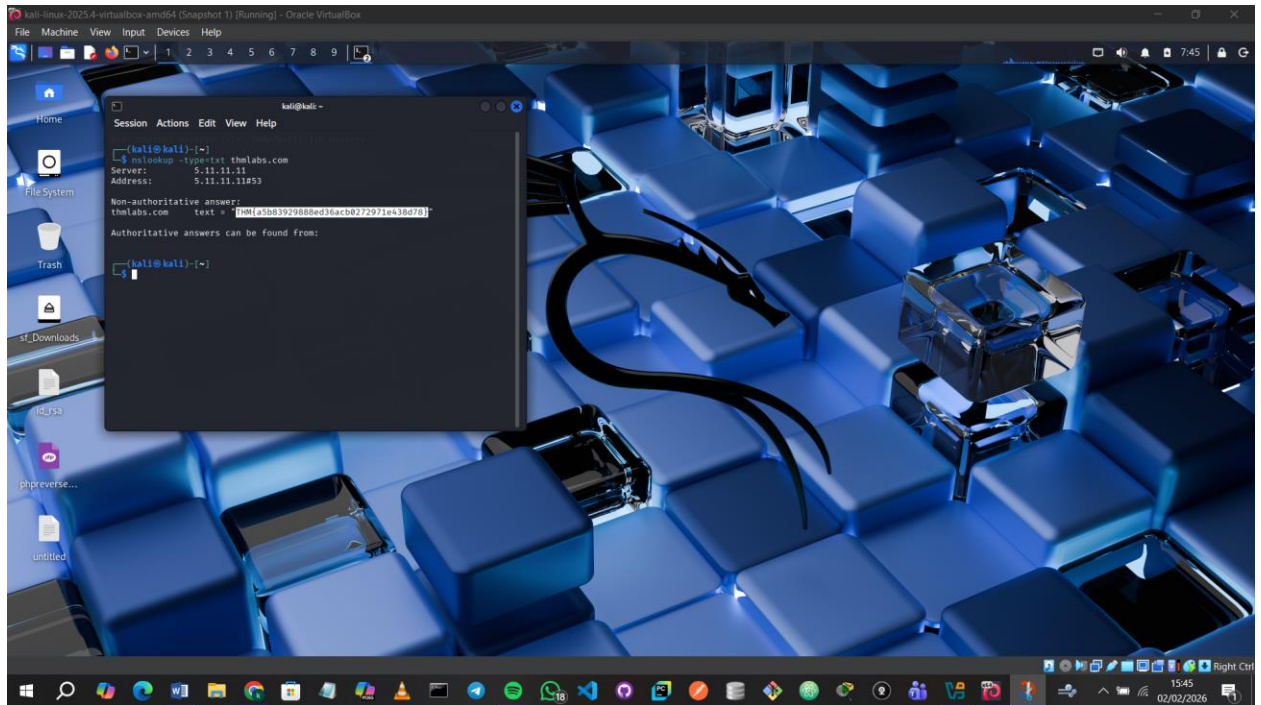


3. nslookup and dig

In this section, I learned how to use nslookup and dig to query DNS records such as A, AAAA, and MX during passive reconnaissance. These tools help identify IP addresses, email servers, and other DNS information that can reveal potential attack surfaces. I also learned that dig provides more detailed output, such as TTL values, compared to nslookup.

Question: Check the TXT records of thmlabs.com. What is the flag there?

Answer: I ran **nslookup -type=txt thmlabs.com** and found it as **THM{a5b83929888ed36acb0272971e438d78}**



4. DNSDumpster

In this section, I learned that tools like nslookup and dig cannot discover subdomains by themselves. I learned how DNSDumpster can be used to identify publicly known subdomains, DNS records, IP addresses, and related server information in a single query. This helps uncover hidden or potentially vulnerable services during passive reconnaissance.

Question: Lookup tryhackme.com on DNSDumpster. What is one interesting subdomain that you would discover in addition to www and blog?

Answer: **remote**

Domain	IP Address	ASN	Location	Hosting Provider	Count
blog.tryhackme.com	104.22.54.228		United States	http: unknown server tech: Cloudflare https: unknown server tech: Cloudflare	5
help.tryhackme.com	172.67.27.10	ASN: 32653 128.0.0.0/2	ENETWORKS, ZA United States	http: unknown server tech: Cloudflare https: unknown server tech: Cloudflare	5
insights-proxy-worker.tryhackme.com	172.67.27.10	ASN: 32653 128.0.0.0/2	ENETWORKS, ZA United States	http: unknown server tech: Cloudflare https: unknown server tech: Cloudflare	5
remote.tryhackme.com	104.22.54.228		United States	http: unknown server tech: Cloudflare https: unknown server tech: Cloudflare	5
www.tryhackme.com	172.67.27.10	ASN: 32653 128.0.0.0/2	ENETWORKS, ZA United States	http: unknown server tech: Cloudflare https: unknown server tech: Cloudflare	5

MX Records					
5 alt1.aspmx.l.google.com	172.253.116.26	ASN: 15169	GOOGLE - Google LLC		
	dj-in-F26.1e100.net	172.253.116.0/24	United States		
5 alt2.aspmx.l.google.com	192.178.223.27	ASN: 15169	GOOGLE - Google LLC		
	galt2-in-F27.1e100.net	192.178.223.0/24	United States		
1 aspmx.l.google.com	142.251.111.26	ASN: 15169	GOOGLE - Google LLC		

5. Shodan.io

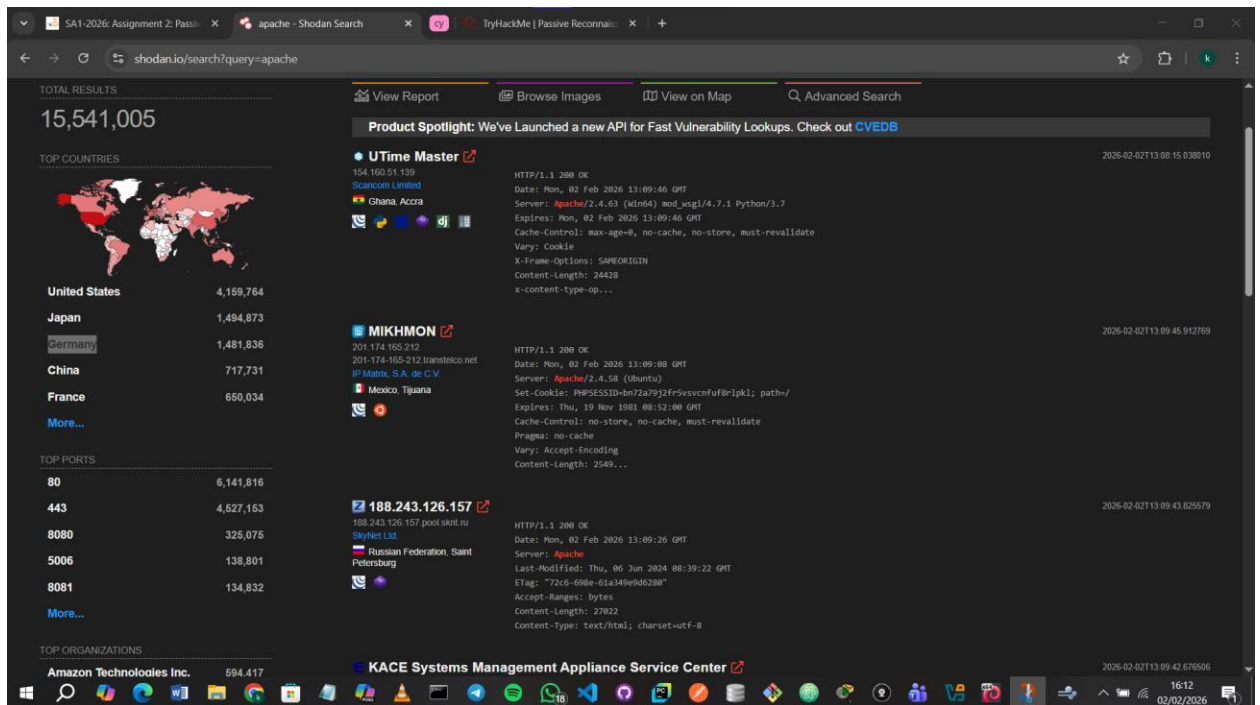
In this section, I learned that Shodan.io is a search engine for internet-connected devices that can be used during passive reconnaissance. It allows me to gather information such as IP addresses, server types, versions, hosting providers, and locations without directly interacting with the target. This is useful for both offensive and defensive security analysis.

Question1: According to Shodan.io, what is the first country in the world in terms of the number of publicly accessible Apache servers?

Answer: I signed in to shodan.io and searched for Apache. The first country was **United States**

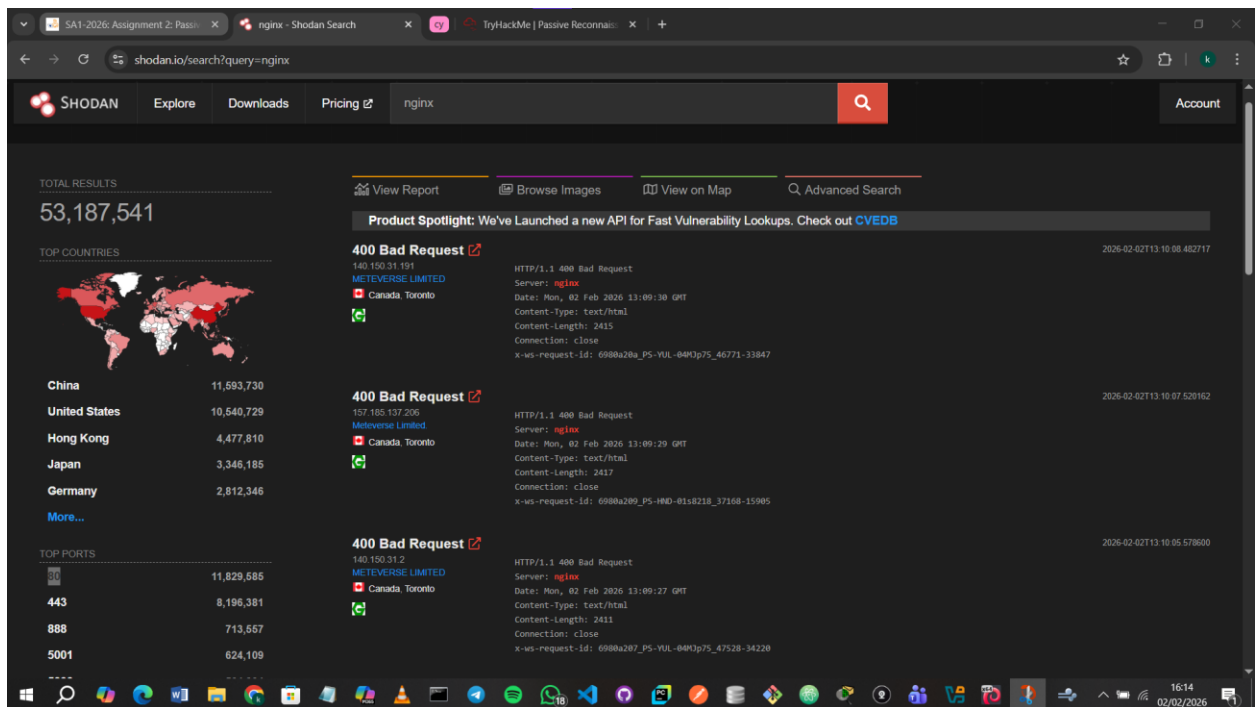
Question2: Based on Shodan.io, what is the 3rd most common port used for Apache?

Answer: **8080**



Question3: Based on Shodan.io, what is the most common port used for nginx?

Answer: I searched for nginx and the port was 80.



6. Summary

A table summary of key techniques and tools learnt in the module:

tryhackme.com/room/passiverecon

Room progress (92%)

In this room, we focused on passive reconnaissance. In particular, we covered command-line tools, `whois`, `nslookup`, and `dig`. We also discussed two publicly available services DNSDumpster and Shodan.io. The power of such tools is that you can collect information about your targets without directly connecting to them. Moreover, the trove of information you may find using such tools can be massive once you master the search options and get used to reading the results.

Purpose	Commandline Example
Lookup WHOIS record	<code>whois tryhackme.com</code>
Lookup DNS A records	<code>nslookup -type=A tryhackme.com</code>
Lookup DNS MX records at DNS server	<code>nslookup -type=MX tryhackme.com 1.1.1.1</code>
Lookup DNS TXT records	<code>nslookup -type=TXT tryhackme.com</code>
Lookup DNS A records	<code>dig tryhackme.com A</code>
Lookup DNS MX records at DNS server	<code>dig @1.1.1.1 tryhackme.com MX</code>
Lookup DNS TXT records	<code>dig tryhackme.com TXT</code>

Learn more about DNS at [DNS in Detail](#).

tryhackme.com/room/passiverecon/congratulations?step=records

You did it! 🎉 Passive Reconnaissance complete!

Points earned	Completed tasks	Streak
🏆 88	✅ 7	🔥 1

Scroll down to review your progress

7. Conclusion

Through this module, I gained a clear understanding of passive reconnaissance and its importance in cybersecurity. I learned how tools such as WHOIS, nslookup, dig, DNSDumpster, and Shodan can be used to collect valuable information while remaining stealthy. This knowledge is essential for identifying potential attack surfaces and understanding an organization's external exposure.